MATH329 · OPERATIONS RESEARCH

Theory Review

The important theory for the midterm in one file

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01 Midterm true/false

Statement	Answer	Midterm
Graphical solutions are only applicable to LP problems with more than three variables.	**F**	472
Operations Research was first initiated during World War I in England to optimize war material utilization.	**F**	472
In a linear programming problem, the objective function or the constraints must be linear.	**F**	472
The Simplex tableau is used to organize the objective function and constraint equations for the algorithm.	**T**	472
The solution of the LPP may be compatible with its constraints imposed by the available resources.	**F**	472 471
The Simplex Algorithm requires the coefficients of the starting basic variables to form an identity matrix.	**T**	472
In Simplex method, if no variable can enter the basis, the problem has no feasible solution.	**F ***	472
All the parameters (coefficients of the objective function and constraints) of the model are unknown.	**F**	471
A basic feasible solution is a feasible solution to LPP which is also the basic solution.	**T**	471
A basic feasible solution is called degenerate if value of at least one basic variable is zero.	**T**	471

If all the ratios in simplex table are negative or infinite then the solution an alternative.	**F**	471

* The 472 answer key shows T. By the lecture, if no variable can enter, the current solution is optimal (the last solution is optimal), so the answer is F.

The instructor's style

He takes a slide sentence and changes one word. Watch these words:

Word in the question	Correct
World War I	World War II
objective **or** constraints	objective **and all** constraints
may be compatible	**must** satisfy
parameters are **unknown**	parameters are **known**
more than three variables	two variables
ratios negative or infinite → **alternative**	**unbounded**
no variable can enter → **no feasible solution**	**optimal**

02 Short theory questions in the midterms

Question	Answer	Midterm
The formal activities of OR were initiated in _____ during World War II.	**England**	461
An optimization problem is called _____ when the objective function and all the constraints are linear.	**LPP**	461
What does the Standard Form mean?	1. All decision variables are nonnegative. 2. All constraints are equations with nonnegative RHS.	461
Mention the special cases of the graphical solution.	1. Alternative optimal solutions 2. Unbounded solution 3. No feasible solution	462
What are the two requirements to start simplex?	1. The basic variables form an identity matrix. 2. Their coefficients in the objective row are zero.	462
What methods can be applied to solve the LPP? (two variables)	Graphical method and Algebraic method.	462 461

03 L1: introduction

Term	Meaning
OR	Started in England during World War II, for the best use of war material. Today it is a decision-making tool.
Model components	Decision variables, then objective function, then constraints.
LPP	The objective and **all** the constraints are linear, with $x_j \geq 0$.
Proportionality	A variable's contribution is proportional to its value.
Additivity	The total is the sum of the variables' contributions.
Feasible solution	Satisfies **all** the constraints.
Optimum feasible solution	The feasible one with the best objective value.
Study phases	Definition, then construction, solution, validation and implementation.
The model alone	Not enough. Intangible factors such as human behaviour count too.

04 L2: graphical method

Term	Meaning
When to use it	Two variables only.
The two steps	Determine the feasible solution space, then the optimum solution.
Where the optimum is	Always at a corner point.
Alternative optima	The objective is parallel to a constraint; infinitely many optima.
Unbounded	The region is open in the direction that improves z.
No feasible solution	The constraint regions do not overlap.
An open region	Does not always mean unbounded (Feed Mix).

05 L3: standard form

Term	Meaning
Standard form	All variables `≥ 0`, all constraints equations with a nonnegative RHS.
Slack	Added `+s` to a `≤` constraint.
Surplus	Subtracted `−s` from a `≥` constraint.
Unrestricted	`x = x⁺ − x⁻` everywhere, including the objective.
Basic solution	Set `n − m` variables to zero (nonbasic) and solve for the rest (basic).
BFS	A basic solution with no negative values. Every BFS is a corner.
Degenerate	At least one basic variable equals zero.
The maximum number	`C(n, m)`, for example `C(4, 2) = 6`.

06 L4: simplex

Term	Meaning
Two starting requirements	The basic variables form an identity matrix, and their objective-row coefficients are zero.
Optimality condition	Entering: max the most negative, min the most positive.
Feasibility condition	Leaving: the smallest ratio, positive coefficients only.
Stopping	No entering variable means the current solution is optimal.
Degeneracy	A tie in the ratio test; a basic variable becomes zero; cycling may occur.
Alternative optima	A nonbasic variable has a zero Z-row coefficient in the optimal tableau.
Unbounded	A variable can enter, but its column is all zero or negative.
Infeasible	The constraints are inconsistent. Never happens when all are $\leq$ with a nonnegative RHS.

07 L5: Big-M

Term	Meaning
Artificial variable	`R` is added to `≥` and `=` constraints to complete the identity matrix.
The penalty	Max `−M`, min `+M`; `M` is a very large positive number.
Fixing the Z row	Min: add `M` × the `R` rows. Max: subtract.
A tie for entering	Prefer the non-artificial variable.
A tie for leaving	Prefer `R` to leave.
R left in the final tableau	At a nonzero level means no feasible solution.

08 Check yourself

1. OR started in the United States during World War II.
2. The graphical method can solve problems with two variables.
3. A surplus variable is added to a ≤ constraint.
4. Every basic feasible solution corresponds to a corner point.
5. For a min problem, the entering variable has the most negative Z-row coefficient.
6. A tie in the minimum ratio test leads to degeneracy.
7. In standard form, every right-hand side must be nonnegative.
8. In the Big-M method, artificial variables get +M in a max problem.
9. An open feasible region always means an unbounded solution.
10. The objective function and all the constraints of an LPP must be linear.

#	Answer	Why
1	**F**	In England.
2	**T**	Two variables only.
3	**F**	A surplus is subtracted from `≥`.
4	**T**	Every BFS is a corner.
5	**F**	For min, the most positive.
6	**T**	And possibly cycling.
7	**T**	Multiply by `−1` if negative.
8	**F**	Max takes `−M`.
9	**F**	The Feed Mix example.
10	**T**	and all, not or.